

Assignment 1: Querying the Bakery Sales Dataset

In this assignment, you will work with a **bakery sales dataset** hosted in our **PostgreSQL Server** environment. Your task is to write SQL queries that extract insights about sales performance, customer behavior, and data quality.

This is a hands-on assignment designed to help you practice:

- Writing analytical SQL queries
- Working with date and time functions
- Applying aggregation and filtering logic
- Evaluating data quality



Submission Instructions

- Submit a single .sql file with your queries, clearly labeled by question.
- You may also include an optional .pdf with explanations or assumptions, if needed.
- Make sure your queries run without syntax errors on the class database.
- Use comments (--) to briefly describe each query, especially if your logic is non-trivial.



Questions (10 points total)

1 Top 3 Most Sold Items by Year-Month (2 points)

Write a query that reports the **top 3 items** with the highest total quantity sold **for each year-month** in the dataset. Your output should include:

- Year and month
- Item name
- Total quantity sold
- Total revenue for each item
- Number of unique tickets containing the item

✓ *Hint:* Use window functions to rank items by quantity within each month.

2 Tickets with 5 or More Unique Articles (1 point)

Identify all **sales tickets** in **December 2021** that include **5 or more unique articles**. Your output should include:

- Ticket ID
- Number of unique items (articles) in that ticket

✓ *Assumption:* “Unique articles” refers to distinct item types in a ticket, not quantity.

3 Most Popular Hour for Traditional Baguette Sales (2 points)

Determine the **hour of the day** when the **Traditional Baguette** was most frequently purchased during **July** (across all years). Your query should:

- Filter for sales of “Traditional Baguette”
- Group by hour (e.g., 14 for 2 PM)
- Return the hour with the highest quantity sold

✓ *Bonus:* You can break ties by selecting the earliest hour of the day in case of equal sales.

4 Busiest Two-Hour Window for Sales (3 points)

Identify the **two-hour window** (e.g., 14:00–16:00) in which the **highest total quantity** of items were sold, across all dates in the dataset. Your output should include:

- The **time range** of the two-hour window
- The **total quantity sold** and **revenue captured** during that window

✓ *Hints:*

- This question requires reasoning about time intervals that **span across rows**. You may need to group by hour and combine adjacent hours.
- Remember that a two-hour window includes both the current hour and the next one.
- Consider using a **self-join**, **window function**, or **aggregation logic** to combine adjacent hours.

5 Data Quality Checks (2 points)

Write queries to assess the **quality of the dataset**. Consider checks for:

- Missing values (e.g., NULLs in important columns)
- Duplicate records
- Outliers (e.g., negative quantities or unusually high values)

Summarize your findings using SQL queries and comments. You may optionally include a few lines of written explanation in your .sql file or in a separate document.